D01 | XOR decision block

All-case aligned observations: 11,981 | Train: 8,491 | Held-out test: 3,490

Test accuracy 0.793 | Majority baseline 0.793 | Balanced accuracy 0.500 | Macro-F1 0.442

B1 -> M08 | aligned support 9,532
B2 -> M01 | aligned support 2,449

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
c.AppType != "Limit raise" AND e.NumberOfTerms <= 41.5 AND c.RequestedAmount <= 12968.5 -> B1 [train n=1,125]

c.AppType != "Limit raise" AND e.NumberOfTerms <= 41.5 AND c.RequestedAmount > 12968.5 -> B1 [train n=298]

c.AppType != "Limit raise" AND e.NumberOfTerms > 41.5 AND c.LoanGoal != "Not speficied" -> B1 [train n=6,221]

c.AppType != "Limit raise" AND e.NumberOfTerms > 41.5 AND c.LoanGoal = "Not speficied" -> B1 [train n=385]

c.AppType = "Limit raise" AND c.LoanGoal != "Other, see explanation" AND e.OfferedAmount <= 8900 -> B1 [train n=53]

c.AppType = "Limit raise" AND c.LoanGoal != "Other, see explanation" AND e.OfferedAmount > 8900 -> B1 [train n=341]

c.AppType = "Limit raise" AND c.LoanGoal = "Other, see explanation" -> B1 [train n=68]
```

Branches never predicted by this tree: B2

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.

Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.

Statistical associations; not bank policy. Structural replay coverage 80.43% is measured without these guards.

D02 | XOR decision block

All-case aligned observations: 21,621 | Train: 15,029 | Held-out test: 6,592

Test accuracy 0.722 | Majority baseline 0.722 | Balanced accuracy 0.500 | Macro-F1 0.419

B1 -> A_Validating | aligned support 15,583

B2 -> M02 | aligned support 6,038

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.FirstWithdrawal <= 0.82499999 AND e.CreditScore <= 310 AND c.AppType != "New credit" -> B1 [train n=645]

e.FirstWithdrawal <= 0.82499999 AND e.CreditScore <= 310 AND c.AppType = "New credit" -> B1 [train n=1,046]

e.FirstWithdrawal <= 0.82499999 AND e.CreditScore > 310 AND c.LoanGoal != "Existing loan takeover" -> B1 [train
n=1,144]

e.FirstWithdrawal <= 0.82499999 AND e.CreditScore > 310 AND c.LoanGoal = "Existing loan takeover" -> B1 [train n=475]

e.FirstWithdrawal > 0.82499999 AND e.NumberOfTerms <= 120.5 AND e.CreditScore <= 926.5 -> B1 [train n=7,911]

e.FirstWithdrawal > 0.82499999 AND e.NumberOfTerms <= 120.5 AND e.CreditScore > 926.5 -> B1 [train n=3,057]

e.FirstWithdrawal > 0.82499999 AND e.NumberOfTerms > 120.5 AND e.NumberOfTerms <= 148 -> B1 [train n=623]

e.FirstWithdrawal > 0.82499999 AND e.NumberOfTerms > 120.5 AND e.NumberOfTerms > 148 -> B1 [train n=128]
```

Branches never predicted by this tree: B2

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.

Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.

Statistical associations; not bank policy. Structural replay coverage 80.43% is measured without these guards.

D03 | XOR decision block

All-case aligned observations: 31,509 | Train: 22,056 | Held-out test: 9,453

Test accuracy 0.837 | Majority baseline 0.651 | Balanced accuracy 0.767 | Macro-F1 0.792

B1 -> A_Submitted | aligned support 20,423
B2 -> M04 | aligned support 11,086

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
c.AppType != "Limit raise" AND c.RequestedAmount <= 4500 -> B2 [train n=1,624]

c.AppType != "Limit raise" AND c.RequestedAmount > 4500 AND c.LoanGoal != "Unknown" -> B1 [train n=17,919]

c.AppType != "Limit raise" AND c.RequestedAmount > 4500 AND c.LoanGoal = "Unknown" -> B2 [train n=142]

c.AppType = "Limit raise" -> B2 [train n=2,371]
```

Branches never predicted by this tree: none

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 80.43% is measured without these guards.

D04 | XOR decision block

All-case aligned observations: 31,509 | Train: 22,056 | Held-out test: 9,453

Test accuracy 0.777 | Majority baseline 0.707 | Balanced accuracy 0.825 | Macro-F1 0.765

B1 -> A_Cancelled | aligned support 9,532
B2 -> parallelGateway | aligned support 21,977

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.CreditScore <= 297 AND c.AppType != "Limit raise" AND e.FirstWithdrawal <= 150.5 -> B1 [train n=3,099]

e.CreditScore <= 297 AND c.AppType != "Limit raise" AND e.FirstWithdrawal > 150.5 -> B1 [train n=7,622]

e.CreditScore <= 297 AND c.AppType = "Limit raise" AND e.FirstWithdrawal <= 1 -> B2 [train n=1,496]

e.CreditScore <= 297 AND c.AppType = "Limit raise" AND e.FirstWithdrawal > 1 -> B2 [train n=545]

e.CreditScore > 297 -> B2 [train n=9,294]
```

Branches never predicted by this tree: none

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.

Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.

Statistical associations; not bank policy. Structural replay coverage 80.43% is measured without these guards.

D05 | XOR decision block

All-case aligned observations: 20,423 | Train: 14,272 | Held-out test: 6,151

Test accuracy 0.829 | Majority baseline 0.829 | Balanced accuracy 0.500 | Macro-F1 0.453

B1 -> W_Handle leads | aligned support 3,493

B2 -> M05 | aligned support 16,930

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
c.LoanGoal != "Existing loan takeover" AND c.RequestedAmount <= 9550 AND c.LoanGoal != "Home improvement" -> B2 [train
n=3,129]

c.LoanGoal != "Existing loan takeover" AND c.RequestedAmount <= 9550 AND c.LoanGoal = "Home improvement" -> B2 [train
n=1,065]

c.LoanGoal != "Existing loan takeover" AND c.RequestedAmount > 9550 AND c.LoanGoal != "Not speficied" -> B2 [train
n=6,942]

c.LoanGoal != "Existing loan takeover" AND c.RequestedAmount > 9550 AND c.LoanGoal = "Not speficied" -> B2 [train
n=460]

c.LoanGoal = "Existing loan takeover" AND c.RequestedAmount <= 31727 AND c.RequestedAmount <= 20950 -> B2 [train
n=1,582]

c.LoanGoal = "Existing loan takeover" AND c.RequestedAmount <= 31727 AND c.RequestedAmount > 20950 -> B2 [train n=544]

c.LoanGoal = "Existing loan takeover" AND c.RequestedAmount > 31727 AND c.RequestedAmount <= 52750 -> B2 [train n=410]

c.LoanGoal = "Existing loan takeover" AND c.RequestedAmount > 31727 AND c.RequestedAmount > 52750 -> B2 [train n=140]
```

Branches never predicted by this tree: B1

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.

Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.

Statistical associations; not bank policy. Structural replay coverage 80.43% is measured without these guards.

D06 | XOR decision block

All-case aligned observations: 36,785 | Train: 25,764 | Held-out test: 11,021

Test accuracy 0.858 | Majority baseline 0.858 | Balanced accuracy 0.500 | Macro-F1 0.462

B1 -> M06 | aligned support 5,276
B2 -> parallelGateway | aligned support 31,509

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.CreditScore <= 597.5 AND c.AppType != "Limit raise" AND e.OfferedAmount <= 9650 -> B2 [train n=4,247]

e.CreditScore <= 597.5 AND c.AppType != "Limit raise" AND e.OfferedAmount > 9650 -> B2 [train n=9,789]

e.CreditScore <= 597.5 AND c.AppType = "Limit raise" AND e.FirstWithdrawal <= 225 -> B2 [train n=1,653]

e.CreditScore <= 597.5 AND c.AppType = "Limit raise" AND e.FirstWithdrawal > 225 -> B2 [train n=549]

e.CreditScore > 597.5 AND c.LoanGoal != "Remaining debt home" AND e.CreditScore <= 853.5 -> B2 [train n=2,784]

e.CreditScore > 597.5 AND c.LoanGoal != "Remaining debt home" AND e.CreditScore > 853.5 -> B2 [train n=6,436]

e.CreditScore > 597.5 AND c.LoanGoal = "Remaining debt home" AND e.NumberOfTerms <= 168 -> B2 [train n=237]

e.CreditScore > 597.5 AND c.LoanGoal = "Remaining debt home" AND e.NumberOfTerms > 168 -> B2 [train n=69]
```

Branches never predicted by this tree: B1

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 80.43% is measured without these guards.

D07 | XOR decision block

All-case aligned observations: 43,598 | Train: 30,319 | Held-out test: 13,279

Test accuracy 0.534 | Majority baseline 0.504 | Balanced accuracy 0.535 | Macro-F1 0.533

B1 -> A_Incomplete | aligned support 21,621
B2 -> D14 | aligned support 21,977

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
c.AppType != "New credit" AND c.LoanGoal != "Existing loan takeover" AND e.CreditScore <= 799 -> B2 [train n=2,463]

c.AppType != "New credit" AND c.LoanGoal != "Existing loan takeover" AND e.CreditScore > 799 -> B2 [train n=324]

c.AppType != "New credit" AND c.LoanGoal = "Existing loan takeover" AND c.RequestedAmount <= 19750 -> B2 [train n=146]

c.AppType != "New credit" AND c.LoanGoal = "Existing loan takeover" AND c.RequestedAmount > 19750 -> B2 [train n=125]

c.AppType = "New credit" AND c.LoanGoal != "Remaining debt home" AND e.MonthlyCost <= 200.37 -> B2 [train n=10,393]

c.AppType = "New credit" AND c.LoanGoal != "Remaining debt home" AND e.MonthlyCost > 200.37 -> B1 [train n=15,584]

c.AppType = "New credit" AND c.LoanGoal = "Remaining debt home" AND e.FirstWithdrawal <= 10875 -> B1 [train n=875]

c.AppType = "New credit" AND c.LoanGoal = "Remaining debt home" AND e.FirstWithdrawal > 10875 -> B1 [train n=409]
```

Branches never predicted by this tree: none

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 80.43% is measured without these guards.

D08 | XOR decision block

All-case aligned observations: 21,621 | Train: 15,029 | Held-out test: 6,592

Test accuracy 0.899 | Majority baseline 0.899 | Balanced accuracy 0.500 | Macro-F1 0.473

B1 -> W_Call incomplete files | aligned support 2,141
B2 -> M03 | aligned support 19,480

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.CreditScore <= 597.5 AND e.MonthlyCost <= 251.93501 AND e.FirstWithdrawal <= 11375 -> B2 [train n=2,176]

e.CreditScore <= 597.5 AND e.MonthlyCost <= 251.93501 AND e.FirstWithdrawal > 11375 -> B2 [train n=431]

e.CreditScore <= 597.5 AND e.MonthlyCost > 251.93501 AND c.LoanGoal != "Remaining debt home" -> B2 [train n=2,399]

e.CreditScore <= 597.5 AND e.MonthlyCost > 251.93501 AND c.LoanGoal = "Remaining debt home" -> B2 [train n=131]

e.CreditScore > 597.5 AND e.FirstWithdrawal <= 3000.5 AND e.CreditScore <= 765.5 -> B2 [train n=344]

e.CreditScore > 597.5 AND e.FirstWithdrawal <= 3000.5 AND e.CreditScore > 765.5 -> B2 [train n=2,735]

e.CreditScore > 597.5 AND e.FirstWithdrawal > 3000.5 AND c.LoanGoal != "Home improvement" -> B2 [train n=4,741]

e.CreditScore > 597.5 AND e.FirstWithdrawal > 3000.5 AND c.LoanGoal = "Home improvement" -> B2 [train n=2,072]
```

Branches never predicted by this tree: B1

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 80.43% is measured without these guards.

D09 | XOR decision block

All-case aligned observations: 1,396 | Train: 951 | Held-out test: 445

Test accuracy 0.726 | Majority baseline 0.726 | Balanced accuracy 0.500 | Macro-F1 0.421

B1 -> M16 | aligned support 1,031
B2 -> M11 | aligned support 365

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.OfferedAmount <= 6750 AND e.MonthlyCost <= 110.14 -> B1 [train n=60]

e.OfferedAmount <= 6750 AND e.MonthlyCost > 110.14 -> B1 [train n=76]

e.OfferedAmount > 6750 AND c.AppType != "New credit" -> B1 [train n=79]

e.OfferedAmount > 6750 AND c.AppType = "New credit" AND e.NumberOfTerms <= 44.5 -> B1 [train n=59]

e.OfferedAmount > 6750 AND c.AppType = "New credit" AND e.NumberOfTerms > 44.5 -> B1 [train n=677]
```

Branches never predicted by this tree: B2

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.

Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.

Statistical associations; not bank policy. Structural replay coverage 80.43% is measured without these guards.

D10 | XOR decision block

All-case aligned observations: 39,916 | Train: 27,964 | Held-out test: 11,952

Test accuracy 0.922 | Majority baseline 0.922 | Balanced accuracy 0.500 | Macro-F1 0.480

B1 -> D11 | aligned support 36,785
B2 -> M12 | aligned support 3,131

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.CreditScore <= 605 AND c.AppType != "Limit raise" AND e.FirstWithdrawal <= 0.32499999 -> B1 [train n=4,629]

e.CreditScore <= 605 AND c.AppType != "Limit raise" AND e.FirstWithdrawal > 0.32499999 -> B1 [train n=11,246]

e.CreditScore <= 605 AND c.AppType = "Limit raise" AND e.OfferedAmount <= 13550 -> B1 [train n=530]

e.CreditScore <= 605 AND c.AppType = "Limit raise" AND e.OfferedAmount > 13550 -> B1 [train n=1,689]

e.CreditScore > 605 AND e.FirstWithdrawal <= 9692.8848 AND c.LoanGoal != "Remaining debt home" -> B1 [train n=5,850]

e.CreditScore > 605 AND e.FirstWithdrawal <= 9692.8848 AND c.LoanGoal = "Remaining debt home" -> B1 [train n=203]

e.CreditScore > 605 AND e.FirstWithdrawal > 9692.8848 AND e.MonthlyCost <= 246.635 -> B1 [train n=1,430]

e.CreditScore > 605 AND e.FirstWithdrawal > 9692.8848 AND e.MonthlyCost > 246.635 -> B1 [train n=2,387]
```

Branches never predicted by this tree: B2

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 80.43% is measured without these guards.

D11 | XOR decision block

All-case aligned observations: 36,785 | Train: 25,764 | Held-out test: 11,021

Test accuracy 0.542 | Majority baseline 0.510 | Balanced accuracy 0.545 | Macro-F1 0.535

B1 -> M10 | aligned support 17,853

B2 -> W_Complete application | aligned support 18,932

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
c.RequestedAmount <= 4500 AND c.AppType != "Limit raise" AND e.OfferedAmount <= 10650 -> B2 [train n=564]

c.RequestedAmount <= 4500 AND c.AppType != "Limit raise" AND e.OfferedAmount > 10650 -> B2 [train n=1,357]

c.RequestedAmount <= 4500 AND c.AppType = "Limit raise" AND e.OfferedAmount <= 12250 -> B2 [train n=113]

c.RequestedAmount <= 4500 AND c.AppType = "Limit raise" AND e.OfferedAmount > 12250 -> B2 [train n=446]

c.RequestedAmount > 4500 AND e.FirstWithdrawal <= 10697 AND e.FirstWithdrawal <= 194 -> B2 [train n=6,891]

c.RequestedAmount > 4500 AND e.FirstWithdrawal <= 10697 AND e.FirstWithdrawal > 194 -> B1 [train n=9,834]

c.RequestedAmount > 4500 AND e.FirstWithdrawal > 10697 AND c.RequestedAmount <= 10775 -> B1 [train n=138]

c.RequestedAmount > 4500 AND e.FirstWithdrawal > 10697 AND c.RequestedAmount > 10775 -> B1 [train n=6,421]
```

Branches never predicted by this tree: none

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.

Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.

Statistical associations; not bank policy. Structural replay coverage 80.43% is measured without these guards.

D12 | XOR decision block

All-case aligned observations: 21,977 | Train: 15,290 | Held-out test: 6,687

Test accuracy 0.599 | Majority baseline 0.513 | Balanced accuracy 0.601 | Macro-F1 0.596

B1 -> M17 | aligned support 11,138
B2 -> M13 | aligned support 10,839

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.CreditScore <= 916.5 AND e.OfferedAmount <= 26730 AND e.CreditScore <= 798.5 -> B2 [train n=5,854]

e.CreditScore <= 916.5 AND e.OfferedAmount <= 26730 AND e.CreditScore > 798.5 -> B1 [train n=2,720]

e.CreditScore <= 916.5 AND e.OfferedAmount > 26730 AND c.LoanGoal != "Remaining debt home" -> B2 [train n=2,305]

e.CreditScore <= 916.5 AND e.OfferedAmount > 26730 AND c.LoanGoal = "Remaining debt home" -> B2 [train n=110]

e.CreditScore > 916.5 AND e.OfferedAmount <= 25050 AND c.LoanGoal != "Remaining debt home" -> B1 [train n=3,498]

e.CreditScore > 916.5 AND e.OfferedAmount <= 25050 AND c.LoanGoal = "Remaining debt home" -> B2 [train n=85]

e.CreditScore > 916.5 AND e.OfferedAmount > 25050 AND e.FirstWithdrawal <= 8266 -> B2 [train n=319]

e.CreditScore > 916.5 AND e.OfferedAmount > 25050 AND e.FirstWithdrawal > 8266 -> B2 [train n=399]
```

Branches never predicted by this tree: none

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 80.43% is measured without these guards.

D13 | XOR decision block

All-case aligned observations: 41,333 | Train: 28,952 | Held-out test: 12,381

Test accuracy 0.890 | Majority baseline 0.890 | Balanced accuracy 0.500 | Macro-F1 0.471

B1 -> M09 | aligned support 4,548
B2 -> M12 | aligned support 36,785

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.CreditScore <= 594.5 AND c.AppType != "Limit raise" AND e.FirstWithdrawal <= 20.825 -> B2 [train n=4,961]

e.CreditScore <= 594.5 AND c.AppType != "Limit raise" AND e.FirstWithdrawal > 20.825 -> B2 [train n=11,801]

e.CreditScore <= 594.5 AND c.AppType = "Limit raise" AND e.FirstWithdrawal <= 1508 -> B2 [train n=1,815]

e.CreditScore <= 594.5 AND c.AppType = "Limit raise" AND e.FirstWithdrawal > 1508 -> B2 [train n=441]

e.CreditScore > 594.5 AND c.AppType != "New credit" AND c.RequestedAmount <= 2650 -> B2 [train n=89]

e.CreditScore > 594.5 AND c.AppType != "New credit" AND c.RequestedAmount > 2650 -> B2 [train n=249]

e.CreditScore > 594.5 AND c.AppType = "New credit" AND c.LoanGoal != "Remaining debt home" -> B2 [train n=9,267]

e.CreditScore > 594.5 AND c.AppType = "New credit" AND c.LoanGoal = "Remaining debt home" -> B2 [train n=329]
```

Branches never predicted by this tree: B1

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 80.43% is measured without these guards.

D14 | XOR decision block

All-case aligned observations: 21,977 | Train: 15,290 | Held-out test: 6,687

Test accuracy 0.857 | Majority baseline 0.780 | Balanced accuracy 0.612 | Macro-F1 0.545

B1 -> A_Cancelled | aligned support 1,031
B2 -> A_Denied | aligned support 3,718
B3 -> O_Accepted | aligned support 17,228

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.CreditScore <= 297 AND c.AppType != "New credit" AND e.NumberOfTerms <= 77.5 -> B3 [train n=353]

e.CreditScore <= 297 AND c.AppType != "New credit" AND e.NumberOfTerms > 77.5 -> B3 [train n=1,255]

e.CreditScore <= 297 AND c.AppType = "New credit" AND e.OfferedAmount <= 8950 -> B2 [train n=1,248]

e.CreditScore <= 297 AND c.AppType = "New credit" AND e.OfferedAmount > 8950 -> B2 [train n=3,140]

e.CreditScore > 297 -> B3 [train n=9,294]
```

Branches never predicted by this tree: B1

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.

Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.

Statistical associations; not bank policy. Structural replay coverage 80.43% is measured without these guards.

D15 | XOR decision block

All-case aligned observations: 17,228 | Train: 12,009 | Held-out test: 5,219

Test accuracy 0.846 | Majority baseline 0.743 | Balanced accuracy 0.704 | Macro-F1 0.742

B1 -> M14 | aligned support 4,436
B2 -> M15 | aligned support 12,792

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.CreditScore <= 596 AND c.AppType != "New credit" AND e.OfferedAmount <= 25150 -> B2 [train n=901]

e.CreditScore <= 596 AND c.AppType != "New credit" AND e.OfferedAmount > 25150 -> B2 [train n=486]

e.CreditScore <= 596 AND c.AppType = "New credit" AND e.NumberOfTerms <= 24.5 -> B1 [train n=50]

e.CreditScore <= 596 AND c.AppType = "New credit" AND e.NumberOfTerms > 24.5 -> B1 [train n=1,283]

e.CreditScore > 596 AND e.NumberOfTerms <= 60.5 AND e.OfferedAmount <= 5050 -> B2 [train n=797]

e.CreditScore > 596 AND e.NumberOfTerms <= 60.5 AND e.OfferedAmount > 5050 -> B2 [train n=2,974]

e.CreditScore > 596 AND e.NumberOfTerms > 60.5 AND c.AppType != "Limit raise" -> B2 [train n=5,232]

e.CreditScore > 596 AND e.NumberOfTerms > 60.5 AND c.AppType = "Limit raise" -> B2 [train n=286]
```

Branches never predicted by this tree: none

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 80.43% is measured without these guards.

D16 | XOR decision block

All-case aligned observations: 5,802 | Train: 4,076 | Held-out test: 1,726

Test accuracy 0.777 | Majority baseline 0.777 | Balanced accuracy 0.500 | Macro-F1 0.437

B1 -> M14 | aligned support 1,366
B2 -> M15 | aligned support 4,436

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
c.AppType != "New credit" AND e.MonthlyCost <= 340 AND c.RequestedAmount <= 10250 -> B2 [train n=50]

c.AppType != "New credit" AND e.MonthlyCost <= 340 AND c.RequestedAmount > 10250 -> B2 [train n=80]

c.AppType != "New credit" AND e.MonthlyCost > 340 -> B2 [train n=61]

c.AppType = "New credit" AND e.OfferedAmount <= 6350 AND e.MonthlyCost <= 117.51 -> B2 [train n=206]

c.AppType = "New credit" AND e.OfferedAmount <= 6350 AND e.MonthlyCost > 117.51 -> B2 [train n=156]

c.AppType = "New credit" AND e.OfferedAmount > 6350 AND e.FirstWithdrawal <= 40557.5 -> B2 [train n=3,430]

c.AppType = "New credit" AND e.OfferedAmount > 6350 AND e.FirstWithdrawal > 40557.5 -> B2 [train n=93]
```

Branches never predicted by this tree: B1

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.

Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.

Statistical associations; not bank policy. Structural replay coverage 80.43% is measured without these guards.

D17 | XOR decision block

All-case aligned observations: 15,697 | Train: 11,037 | Held-out test: 4,660

Test accuracy 0.702 | Majority baseline 0.699 | Balanced accuracy 0.519 | Macro-F1 0.466

B1 -> M13 | aligned support 4,858

B2 -> M17 | aligned support 10,839

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
c.LoanGoal != "Remaining debt home" AND e.CreditScore <= 882.5 AND e.CreditScore <= 597.5 -> B2 [train n=4,817]

c.LoanGoal != "Remaining debt home" AND e.CreditScore <= 882.5 AND e.CreditScore > 597.5 -> B2 [train n=2,831]

c.LoanGoal != "Remaining debt home" AND e.CreditScore > 882.5 AND e.OfferedAmount <= 25800 -> B2 [train n=1,922]

c.LoanGoal != "Remaining debt home" AND e.CreditScore > 882.5 AND e.OfferedAmount > 25800 -> B2 [train n=825]

c.LoanGoal = "Remaining debt home" AND e.FirstWithdrawal <= 10125 AND e.FirstWithdrawal <= 3051.535 -> B1 [train
n=347]

c.LoanGoal = "Remaining debt home" AND e.FirstWithdrawal <= 10125 AND e.FirstWithdrawal > 3051.535 -> B1 [train n=103]

c.LoanGoal = "Remaining debt home" AND e.FirstWithdrawal > 10125 AND e.MonthlyCost <= 264.98001 -> B2 [train n=102]

c.LoanGoal = "Remaining debt home" AND e.FirstWithdrawal > 10125 AND e.MonthlyCost > 264.98001 -> B2 [train n=90]
```

Branches never predicted by this tree: none

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.

Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.

Statistical associations; not bank policy. Structural replay coverage 80.43% is measured without these guards.

D18 | XOR decision block

All-case aligned observations: 4,693 | Train: 3,249 | Held-out test: 1,444

Test accuracy 0.793 | Majority baseline 0.793 | Balanced accuracy 0.500 | Macro-F1 0.442

B1 -> M18 | aligned support 975
B2 -> M16 | aligned support 3,718

Learned rules (training cases only; depth <= 3; minimum leaf size 50)

```
e.OfferedAmount <= 11950 AND e.FirstWithdrawal <= 1240 AND c.RequestedAmount <= 5950 -> B2 [train n=163]

e.OfferedAmount <= 11950 AND e.FirstWithdrawal <= 1240 AND c.RequestedAmount > 5950 -> B2 [train n=145]

e.OfferedAmount <= 11950 AND e.FirstWithdrawal > 1240 AND e.FirstWithdrawal <= 8725 -> B2 [train n=917]

e.OfferedAmount <= 11950 AND e.FirstWithdrawal > 1240 AND e.FirstWithdrawal > 8725 -> B2 [train n=244]

e.OfferedAmount > 11950 AND c.AppType != "Limit raise" AND c.RequestedAmount <= 11400 -> B2 [train n=227]

e.OfferedAmount > 11950 AND c.AppType != "Limit raise" AND c.RequestedAmount > 11400 -> B2 [train n=1,408]

e.OfferedAmount > 11950 AND c.AppType = "Limit raise" AND e.OfferedAmount <= 21250 -> B2 [train n=59]

e.OfferedAmount > 11950 AND c.AppType = "Limit raise" AND e.OfferedAmount > 21250 -> B2 [train n=86]
```

Branches never predicted by this tree: B1

c. = case input at start; e. = last synchronized event input. Missing numeric value = -1; explicit *_known flags.
Inputs update after synchronous moves only. One optimal alignment per variant; alternative alignments may change decisions.
Statistical associations; not bank policy. Structural replay coverage 80.43% is measured without these guards.